COURSE TITLE : OFFSET PRINTING
COURSE CODE : 3104
COURSE CATEGORY : B
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Basic features of sheet fed press	25
2	Single and multicolor printing	18
3	Pre make ready and make ready operations	17
4	Trouble shooting and ancillary processes	15
TOTAL		**75**

Course General Outcome:

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To Understand the characteristics of sheet fed offset printing machines
2	To Know different types of sheet-fed offset machine
3	To Comprehend the printing process with sheet-fed offset machine
4	To Understand web offset printing
5	To Understand the troubleshooting in offset printing.

MODULE I BASIC FEATURES OF SHEET FED PRESS

1.1.0 To understand the characteristics of sheet fed printing machines

1.1.1. To classify offset printing machines
1.1.2. To list out major national and international Printing press manufactures
1.1.3. To explain about different units and sub units of sheet fed offset machine
1.1.4. To describe mechanical and operational features of sheet fed offset machine
1.1.5. To explain anti set off methods

1.2.0 To Know different types of sheet-fed offset machine

1.2.1. To identify different types of sheet fed offset machine
1.2.2. To define perfecting sheet fed press

MODULE II SINGLE AND MULTICOLOR PRINTING

2.1.0 To Comprehend the printing process with sheet-fed offset machine

2.1.1 To explain the printing process with single color sheet fed offset machine
2.1.2 To explain the printing process with multicolor sheet fed printing machine
2.1.3 To identify the sheet travel during perfecting and non perfecting printing

2.1.4 To define sheet reversal system
2.1.5 To differentiate wet on wet and wet on dry printing
2.1.6 To select correct color sequence for the job
2.1.7 To identify features of modern computer controlled offset printing machines
2.1.8 To identify coating processes
2.1.9 To identify varnishing operations
2.1.10 To define pre makeready and makeready
2.1.11 To identify the purpose and advantage of pre makeready and makeready

MODULE III WEB OFFSET

3.1.0 To Understand web offset printing

3.1.1 To define basic design features of web printing machine
3.1.2 To identify machine configuration
3.1.3 To know the working of splicer mechanism
3.1.4 To know the difference between web printing unit and sheet fed printing unit
3.1.5 To know about drying and chilling rollers
3.1.6 To explain the delivery folding mechanism

MODULE IV TROUBLE SHOOTING

4.1.0 To Understand the troubleshooting in offset printing

4.1.1 To identify the basic printing problems with offset printing
4.1.2 To explain troubles and remedies related to paper
4.1.3 To explain troubles and remedies related to ink
4.1.4 To explain troubles and remedies related to dampening solution
4.1.5 To explain troubles and remedies related to blanket
4.1.6 To explain troubles and remedies related to printing machine

CONTENT DETAILS

MODULE I

Classification of Offset printing machines – Sheet fed, web fed. Introduction to Major offset presses, currently used in industry and their manufacturers (national and international).

<u>Design features and functional parts of a sheet fed offset machine</u> (parts introduction and its functions):

Feeding unit: Types of Feeders – Friction feeders, Suction feeders, Types of Suction Feeders – Single sheet feeder and Stream Feeder. Pile board. Stoppers – front, side, back. Air blower nozzle. Suckers – lifting suckers, forwarding suckers. Flip arrangements – metal, brush.

Feed board assemblies: introduction roller, drop down wheel, rubber wheel, brush wheels, endless tapes, ball arrangement. Modern feed board sheet forwarding.

Registration unit: Front lay. Side lay – push type, pull type. Comparison between push type and pull type lays. Sheet detectors – no-sheet detector, multi sheet detectors, cross sheet detectors. Types of detectors (brief introduction only) – mechanical and photo electrical sheet detectors.

Sheet insertion system/infeed unit: Swing arm, Rotary, Tumbler and Overfeed Grippers.

Printing Unit: Construction and parts of Plate cylinder. Construction and parts of Blanket cylinder. Construction and parts of Impression cylinder. Bearer contact and bearer clearance printing unit.

Inking system – types, construction and features. (pyramid, multi roller) modern inking systems with motorized ink control. Digital inking system.

Dampening system – types, construction and features. Contact and noncontact dampening system (Vibrator-type dampening system; Continuous-type dampening system; Indirect dampening solution application through the ink roller; Brush-type dampening system; Centrifugal dampening system; jet spray dampening system)

Ink water balance (in detail).

Delivery unit : Delivery Assist Devices, Suction Slow down Rollers, Blow downs, Wedges, Skeleton Wheels, Star Wheels.

Anti-setoff devices – need of anti setoff devices. Types of anti-setoff devices.

Types of Sheet fed offset press – single color, multi color (2 color to 8 color). Perfecting presses. Sheet reversal systems.

MODULE II

Explanation of how the sheet travels in the machine during the printing process (in single color and multi color machines).

Printing of single color work with single color sheet fed machine. Printing of multicolor work with single color sheet fed machine. Wet on wet printing and Wet on dry printing.

Sheet reversal systems. Explanation of how the sheet travels in the perfecting sheet fed machine during the printing process.

Multicolor printing - with 4 color machine. Color sequence. Selection of color sequence – factors influencing color sequence. color sequence used in the industry

Computer controlled offset machines and their features – ink setting, sheet control, registration control. Automatic blanket and impression cleaning systems. Automatic plate cleaning system.

Coating – definition, need, process, type of coating-solid, solid with cut out, spot, gold and silver effect.

Varnishing – definition, need, process, types of varnishing.

Makeready and permakeready – definition, need, advantage, process (brief explanation of standard procedure), current industry practice.

MODULE III

Basic design features of web offset printing machine – feeding unit, web control unit, printing unit, drying unit, delivery unit.

Configurations of web offset printing machine – blanket to blanket, inline, vertical, satellite.

Feeding unit – reel stand and its types.

Web Splicer – definition, need, types, working of - Zero speed splicer, Flying splicer.

Web Control – Dancer Roller, Metering Roller, Box Tilt, Web break detectors, Bustle Wheel

Printing unit - Difference between sheet fed printing unit and web fed printing unit.

Drying unit – Heat and cold set. Types of Dryers - Open flame, High Velocity Hot Air and Combination Dryer.

Chill Rollers – Early stage Chill rolls, Baffle plate Chill rolls and Jacketed Chill rolls.

Delivery unit : Folders - Former folder, Double Former Folder, Jaw Folder, Chopper Folder, Combination Folder and Ribbon Folder. Auxiliary Equipments – Stackers, Bundlers, Sheeters, Perforators and Imprinters

MODULE IV

Troubles related to paper their remedies

Troubles related to ink and their remedies

Troubles related to dampening solution and their remedies

Troubles related to blanket and their remedies

Troubles related to printing machine and their remedies

Masters for offset printing. Conventional plate making (introduction only). Types of plate bases – aluminum, poly-master . PS plates. Names of major manufacturers of PS plates.

CTP (computer to plate) technology. Types of CTP plates. Types of CTP systems – UV , Thermal. Working of CTP machine. Names of major manufacturers of CTP machines.

CTcP (computer to conventional plate) technology .

Computer to machine/ computer to press. Direct imaging presses.

Reference :

Author	**Title**	**Publishers**
Dejdas LP and Destree TM	Sheet fed offset press operating	GATF, USA
A S Porter	Manual for lithographic press operation	
C S Misra	Offset technology	Anupam prakashan
Fonaid E Tood	Printing inks	PIRA international united kingdom
W.H. Bureau	What the printer should know about the paper	GATF
Michael Bernard, John Peacock.	Handbook of printing and production	Springer Science & Business Media
Heigh. M. Speir	Introduction in Printing Technology	
Adams J.M, Faux,D.D and Rieber L.J	Printing Technology	Delmar Publishers, NewYork
Helmut Kipphan	Hand book of print media	Springer Science & Business Media
GATF staff	Solving sheet fed offset press problems	GATF, USA